Graph Theory & Networks (AI HL)

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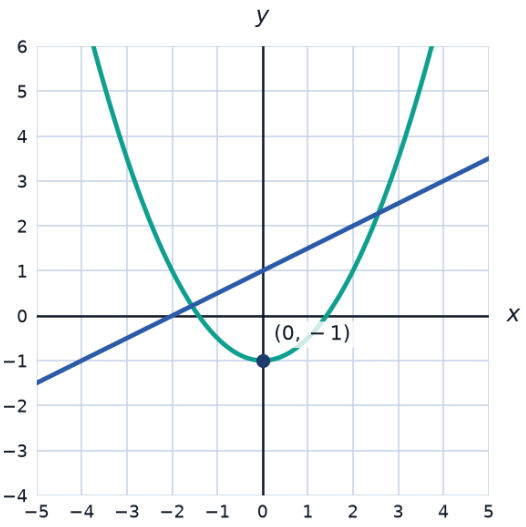
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Degrees and the Handshaking Lemma**

A weighted graph has edges (with weights) $1–2(1), 2–3(6), 3–4(5), 4–5(7), 1–5(8), 2–5(2), 1–3(9)$. Find the total weight of the minimum spanning tree.

2 **[Foundational]** • **Target: Eulerian and Hamiltonian Paths and Circuits**

A weighted graph has edges (with weights) $1–2(9), 2–3(5), 3–4(7), 4–5(2), 1–5(3), 2–5(4), 1–3(8)$. Find the total weight of the minimum spanning tree.



3 **[Foundational]** • **Target: Minimum Spanning Trees (Kruskal & Prim)**

A weighted graph has edges (with weights) $1–2(8), 2–3(6), 3–4(3), 4–5(1), 1–5(4), 2–5(7), 1–3(9)$. Find the total weight of the minimum spanning tree.

4 **[Foundational]** • **Target: Shortest Paths and Dijkstra's Algorithm**

A weighted graph has edges (with weights) $1–2(3), 2–3(1), 3–4(2), 4–5(4), 1–5(5), 2–5(6), 1–3(9)$. Find the total weight of the minimum spanning tree.

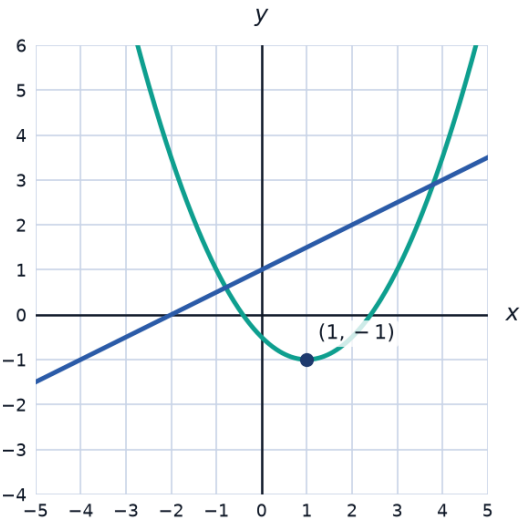
5 **[Foundational]** • **Target: Degrees and the Handshaking Lemma**

A weighted graph has edges (with weights) $1–2(1), 2–3(6), 3–4(3), 4–5(2), 1–5(4), 2–5(5), 1–3(8)$. Find the total weight of the minimum spanning tree.

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6 [Foundational] • Target: Eulerian and Hamiltonian Paths and Circuits

A weighted graph has edges (with weights) $1{-}2(1), 2{-}3(4), 3{-}4(3), 4{-}5(5), 1{-}5(6), 2{-}5(7), 1{-}3(8)$. Find the total weight of the minimum spanning tree.



7 [Foundational] • Target: Minimum Spanning Trees (Kruskal & Prim)

A weighted graph has edges (with weights) $1{-}2(6), 2{-}3(8), 3{-}4(2), 4{-}5(4), 1{-}5(7), 2{-}5(9), 1{-}3(10)$. Find the total weight of the minimum spanning tree.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Shortest Paths and Dijkstra's Algorithm

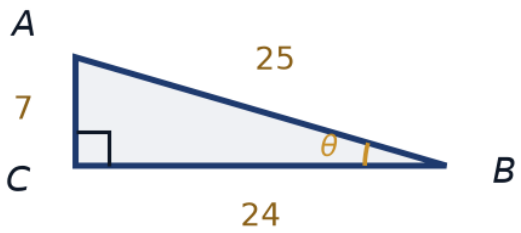
Consider the function $f(x) = \frac{1}{2}x^2 - 3x + 5$. The graph of $y = f(x)$ is a parabola. Find the x-coordinate of the vertex of the parabola.

9 [Intermediate] • Target: Degrees and the Handshaking Lemma

State the principal value (in degrees, $-90^\circ \le \theta \le 90^\circ$) satisfying $\sin\theta = -\frac{1}{2}$.

10 [Intermediate] • Target: Eulerian and Hamiltonian Paths and Circuits

If $z = 2(\cos 45^\circ + i\sin 45^\circ)$, find $|z^6|$ and $\arg(z^6)$ (in degrees, $0 \le \arg < 360$).



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11 [Intermediate] • Target: Minimum Spanning Trees (Kruskal & Prim)

If $z = 2(\cos 30^\circ + i\sin 30^\circ)$, find $|z^6|$ and $\arg(z^6)$ (in degrees, $0 \le \arg < 360$).

12 [Intermediate] • Target: Shortest Paths and Dijkstra's Algorithm

If $z = 1(\cos 60^\circ + i\sin 60^\circ)$, find $|z^3|$ and $\arg(z^3)$ (in degrees, $0 \le \arg < 360$).

13 [Intermediate] • Target: Degrees and the Handshaking Lemma

If $z = 1(\cos 90^\circ + i\sin 90^\circ)$, find $|z^6|$ and $\arg(z^6)$ (in degrees, $0 \le \arg < 360$).

14 [Intermediate] • Target: Eulerian and Hamiltonian Paths and Circuits

If $z = 2(\cos 60^\circ + i\sin 60^\circ)$, find $|z^3|$ and $\arg(z^3)$ (in degrees, $0 \le \arg < 360$).

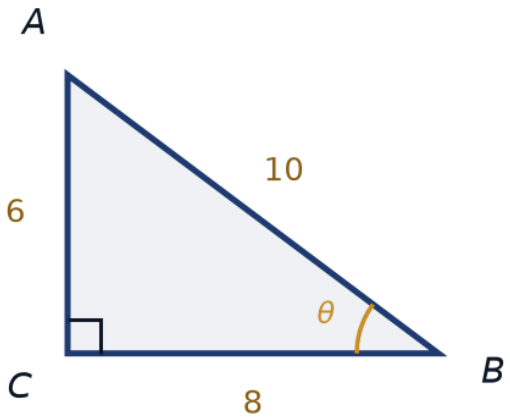
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Minimum Spanning Trees (Kruskal & Prim) • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

If $z = 1(\cos 45^\circ + i\sin 45^\circ)$, find $|z^5|$ and $\arg(z^5)$ (in degrees, $0 \le \arg < 360$).

16 [Challenging] • Target: Shortest Paths and Dijkstra's Algorithm • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

If $z = 2(\cos 60^\circ + i\sin 60^\circ)$, find $|z^6|$ and $\arg(z^6)$ (in degrees, $0 \le \arg < 360$).



17 [Challenging] • Target: Degrees and the Handshaking Lemma • Cross-Lesson: 6.01 Matrices (AI HL)

If $z = 2(\cos 30^\circ + i\sin 30^\circ)$, find $|z^3|$ and $\arg(z^3)$ (in degrees, $0 \le \arg < 360$).

18 [Challenging] • Target: Eulerian and Hamiltonian Paths and Circuits • Cross-Lesson: 6.03 Markov Chains (AI HL)

If $z = 1(\cos 60^\circ + i\sin 60^\circ)$, find $|z^6|$ and $\arg(z^6)$ (in degrees, $0 \le \arg < 360$).

19 [Challenging] • Target: Minimum Spanning Trees (Kruskal & Prim) • Cross-Lesson: 1.01 Proof by Mathematical Induction

A graph has vertices $\{1, \ldots, 6\}$ and edges $1{-}5, 1{-}4, 4{-}5, 2{-}6, 2{-}5, 1{-}3$. Find the degree of vertex 2.

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20 [Challenging] • Target: Shortest Paths and Dijkstra's Algorithm • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

A graph has vertices $\{1, \ldots, 6\}$ and edges $1{-}2, 5{-}6, 2{-}5, 1{-}3, 3{-}5$. Find the degree of vertex 2.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Degrees and the Handshaking Lemma

What does the Handshaking Lemma state for any undirected graph with V vertices and E edges?

a. The sum of the degrees of all vertices equals twice the number of edges: $\sum \deg(v) = 2E$

b. The sum of the degrees equals E

c. The number of vertices with odd degree is always odd

d. Every vertex must have degree at least 2

22 [Foundational] • Target: Degrees and the Handshaking Lemma

A graph has 6 vertices, each having degree 3. How many edges does the graph have?

a. 9

b. 18

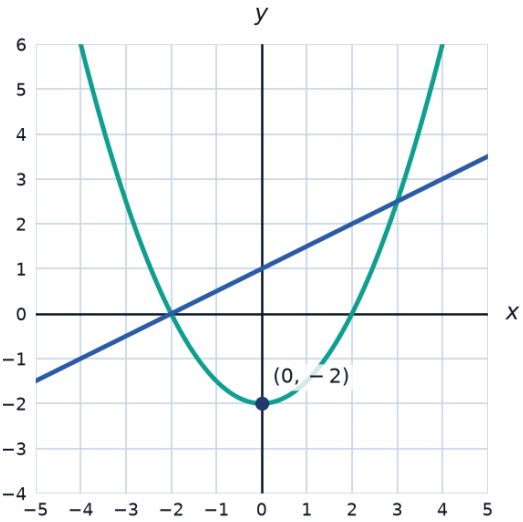
c. 6

d. 12

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23 [Foundational] • Target: Degrees and the Handshaking Lemma

Why can a graph never have an odd number of vertices of odd degree?



a. Because $\sum \deg(v) = 2E$ is always even, the sum of odd degrees must be even, forcing an even count of odd vertices

b. Because edges cannot cross each other

c. By Euler's formula $V - E + F = 2$

d. Because every vertex must have at least one adjacent edge

24 [Foundational] • Target: Eulerian and Hamiltonian Paths and Circuits

What condition is both necessary and sufficient for a connected graph to contain an Eulerian circuit (a closed trail visiting every edge exactly once)?

a. Every vertex has an even degree

b. Exactly two vertices have odd degree

c. The graph contains no cycles

d. The graph is a complete bipartite graph

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25 [Foundational] • Target: Eulerian and Hamiltonian Paths and Circuits

If a connected graph has exactly two vertices of odd degree, what type of Eulerian walk does it contain?

a. An Eulerian trail (open path visiting every edge once) that starts at one odd vertex and ends at the other

b. An Eulerian circuit returning to the start

c. No Eulerian trail or circuit can exist

d. A Hamiltonian cycle

26 [Foundational] • Target: Eulerian and Hamiltonian Paths and Circuits

What is the difference between an Eulerian circuit and a Hamiltonian cycle?

a. An Eulerian circuit visits every edge once; a Hamiltonian cycle visits every vertex once

b. An Eulerian circuit visits every vertex once; a Hamiltonian cycle visits every edge once

c. They are identical concepts in graph theory

d. Hamiltonian cycles apply only to directed trees

27 [Foundational] • Target: Minimum Spanning Trees (Kruskal & Prim)

In a connected weighted graph with V vertices, how many edges does any spanning tree contain?

a. $V - 1$

b. V

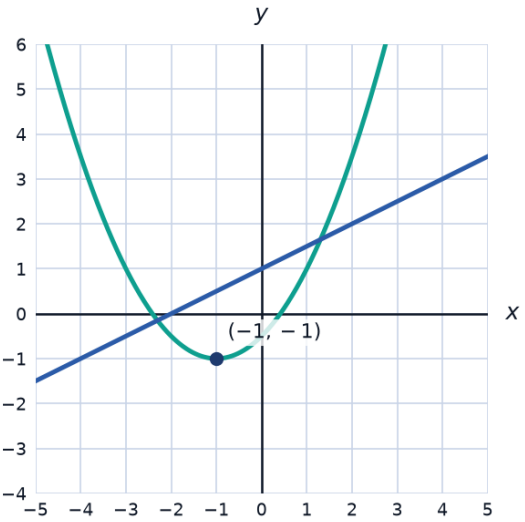
c. $V + 1$

d. $\frac{V(V-1)}{2}$

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28 [Foundational] • Target: Minimum Spanning Trees (Kruskal & Prim)

How does Kruskal's algorithm select edges to construct a Minimum Spanning Tree (MST)?



a. Sorts all edges by weight and greedily adds the smallest available edge that does not create a cycle

b. Grows a single tree by continuously adding the closest vertex to the current tree

c. Removes the largest cycle edges one by one until only a tree remains

d. Visits vertices in alphabetical order

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Minimum Spanning Trees (Kruskal & Prim)

How does Prim's algorithm differ conceptually from Kruskal's algorithm?

a. Prim grows a single connected tree from an arbitrary starting vertex, whereas Kruskal maintains a forest of components until unified

b. Prim works only on unweighted graphs

c. Kruskal finds the longest path rather than minimum tree

d. Prim can only be applied to planar graphs

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30 [Intermediate] • Target: Shortest Paths and Dijkstra's Algorithm

What is the purpose of Dijkstra's algorithm?

a. To find the shortest paths from a single source vertex to all other vertices in a network with non-negative edge weights

b. To find a minimum spanning tree

c. To determine if a graph has an Eulerian circuit

d. To find the maximum flow through a network

31 [Intermediate] • Target: Shortest Paths and Dijkstra's Algorithm

Why does Dijkstra's algorithm fail if a network contains negative edge weights?

a. A greedy choice once finalized could be improved later by a negative edge, invalidating the optimality assumption

b. Distances cannot be negative by definition

c. The algorithm enters an infinite cycle immediately

d. Square roots of negative weights cannot be computed

32 [Intermediate] • Target: Shortest Paths and Dijkstra's Algorithm

In a planar graph, if $V = 6$ and $E = 9$, how many faces F does the graph enclose according to Euler's formula $V - E + F = 2$?

a. 5

b. 3

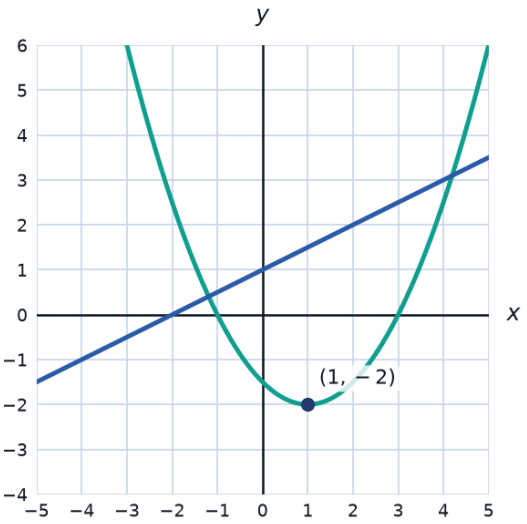
c. 4

d. 6

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33 [Intermediate] • Target: Degrees and the Handshaking Lemma

A graph has vertices $\{1, \ldots, 6\}$ and edges $1{-}5, 2{-}3, 5{-}6, 3{-}6, 1{-}6, 2{-}5$. Find the degree of vertex 5.



a. 3

b. 5

c. 1

d. 4

34 [Intermediate] • Target: Eulerian and Hamiltonian Paths and Circuits

A graph has vertices $\{1, \ldots, 6\}$ and edges $2{-}4, 4{-}6, 2{-}6, 2{-}5, 1{-}3, 3{-}5$. Find the degree of vertex 4.

a. 2

b. 4

c. 0

d. 1

35 [Intermediate] • Target: Minimum Spanning Trees (Kruskal & Prim)

A graph has vertices $\{1, \ldots, 6\}$ and edges $2{-}4, 1{-}2, 1{-}4, 5{-}6, 3{-}6, 2{-}5$. Find the degree of vertex 4.

a. 4

b. 1

c. 0

d. 2

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36 [Intermediate] • Target: Shortest Paths and Dijkstra's Algorithm

A graph has vertices $\{1, \ldots, 5\}$ and edges $1{-}5, 1{-}4, 4{-}5, 2{-}5, 3{-}5$. Find the degree of vertex 4.

a. 2

b. 0

c. 1

d. 4

37 [Intermediate] • Target: Degrees and the Handshaking Lemma

A graph has vertices $\{1, \ldots, 6\}$ and edges $2{-}4, 1{-}2, 4{-}5, 3{-}6, 1{-}6, 1{-}3$. Find the degree of vertex 6.

a. 2

b. 1

c. 4

d. 0

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Eulerian and Hamiltonian Paths and Circuits • Cross-Lesson: 6.03 Markov Chains (AI HL)

A graph has vertices $\{1, \ldots, 5\}$ and edges $2{-}4, 1{-}2, 3{-}4, 1{-}4, 2{-}5$. Find the degree of vertex 2.

a. 3

b. 5

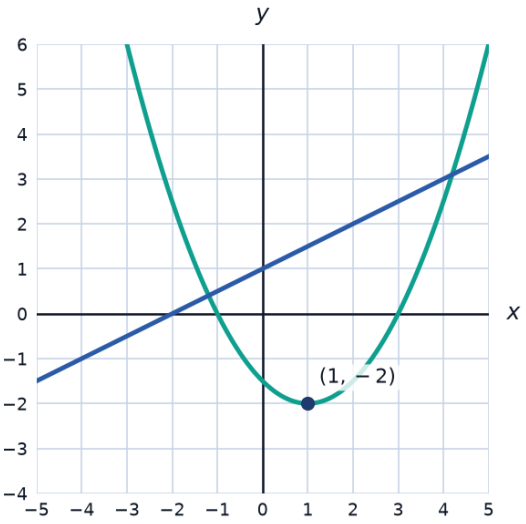
c. 4

d. 2

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39 [Challenging] • Target: Minimum Spanning Trees (Kruskal & Prim) • Cross-Lesson: 1.01 Proof by Mathematical Induction

State the principal value (in degrees, $-90^\circ \le \theta \le 90^\circ$) satisfying $\sin\theta = \frac{1}{2}$.



a. 50°

b. 30°

c. 69°

d. 62°

40 [Challenging] • Target: Shortest Paths and Dijkstra's Algorithm • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

Evaluate $\sin^{-1}\left(-\frac{1}{2}\right)$ in degrees.

a. -30°

b. -24°

c. -32°

d. -17°

41 [Challenging] • Target: Degrees and the Handshaking Lemma • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity

Evaluate $\cos^{-1}\left(-\frac{1}{2}\right)$ in degrees.

a. 120°

b. 97°

c. 142°

d. 143°

42 [Challenging] • Target: Eulerian and Hamiltonian Paths and Circuits • Cross-Lesson: 1.04 Partial Fractions

Evaluate $\sin^{-1}\left(\frac{1}{2}\right)$ in degrees.

a. 41°

b. 30°

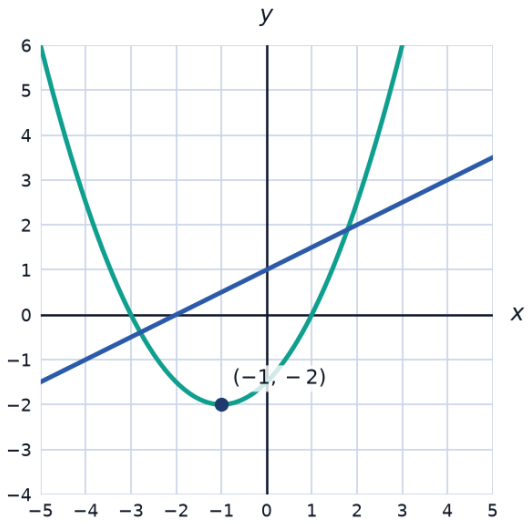
c. 15°

d. 24°

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43 [Challenging] • Target: Minimum Spanning Trees (Kruskal & Prim) • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

State the principal value (in degrees, $-90^\circ \le \theta \le 90^\circ$) satisfying $\sin\theta = 1$.



a. 59°

b. 90°

c. 89°

d. 82°

44 [Challenging] • Target: Shortest Paths and Dijkstra's Algorithm • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

State the principal value (in degrees, $-90^\circ \le \theta \le 90^\circ$) satisfying $\sin\theta = 0$.

a. 16°

b. 0°

c. -34°

d. 18°

45 [Challenging] • Target: Degrees and the Handshaking Lemma • Cross-Lesson: 3.01 3D Vectors & the Cross Product

Evaluate $\tan^{-1}(-1)$ in degrees.

a. -55°

b. -58°

c. -35°

d. -45°

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